

# Advanced Structural Analysis using ANSYS

## Lecture and Exercise Notes

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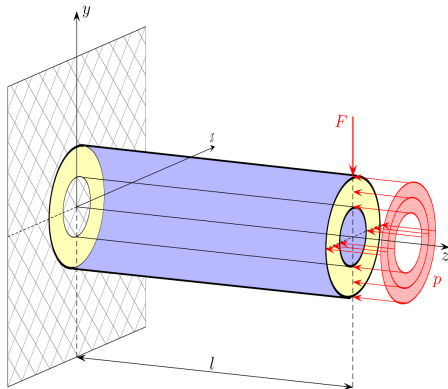
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### Exercise 2: First Problems of Large Deformation

Die Veranstaltung basiert u.a. auf dem Vorlesungs- und Übungsskript „Die Methode der finiten Elemente“, gehalten durch Prof. Dr. rer. nat. M. Braun, Dr.-Ing. H. Walter, Dr.-Ing. U. Zwiers und Dr.-Ing. Philipp Olle, am Lehrstuhl für Mechanik (Maschinenbau) der Universität Duisburg-Essen im WS2004/05

# Cantilever with End Load

## Problem 1



- Geometric data

$$l = 1000 \text{ mm}$$

$$r_a = 80 \text{ mm}$$

$$r_i = 40 \text{ mm}$$

- Material data (plastics)

$$E = 3000 \text{ N/mm}^2$$

$$\nu = 0.2$$

- Clamping

Fixed support at one end of the tube  
Deformation of cross section possible

- Load (cases 1,2,3)

$$F = 6000 \text{ N}$$

$$p_0 = 0 \text{ N/mm}^2, p_1 = 2 \text{ N/mm}^2,$$

$$p_2 = 4 \text{ N/mm}^2$$

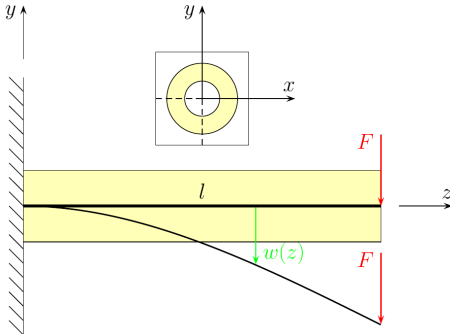
- Objective target

Bending diagram for  $p_0, p_1, p_2$

Deflection of the tube end

# Cantilever with End Load

Beam Theory for  $p_0 = 0 \text{ N/mm}^2$



- Moment of inertia

$$I = \frac{\pi}{4} (r_a^4 - r_i^4)$$

- Bending moment

$$M(z) = F(l - z)$$

- Load

$$F = 6000 \text{ N}$$

## Bending line

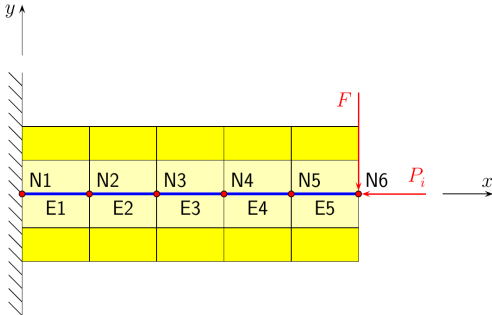
$$w(z) = \frac{2F}{3\pi E(r_a^4 - r_i^4)} (3lz^2 - z^3)$$

## Deflection of the tube end

$$w(l = 1000 \text{ mm}) = 22.1 \text{ mm}$$

# FE-Model Consisting of 5 Beam Elements

Cantilever with End Load



- Element type

Linear finite strain beam  
element BEAM188

- Element properties

$$l_i = 200 \text{ mm}$$

$$h_i = 160 \text{ mm}$$

$$A_i = 15079.6 \text{ mm}^2$$

$$I_i = 30159289.5 \text{ mm}^4$$

- Load (case 1,2,3)

$$F = 6000 \text{ N}$$

$$P_0 = p_0 \cdot A_i = 0 \text{ N}$$

$$P_1 = p_1 \cdot A_i = 30159.3 \text{ N}$$

$$P_2 = p_2 \cdot A_i = 60318.6 \text{ N}$$

- Result  $|u_y|$

Node N1 ... Node N6

# Analysis of Large Deformations

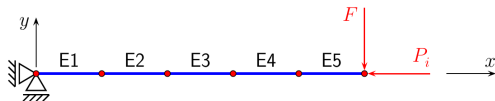
## Cantilever with End Load

### Load step control for nonlinear analyses

#### Small deformation

Bending moment:

$$M(x) = -F(l - x)$$



```
/SOLU
```

```
/com, Boundary conditions
```

```
D,1,all,0
```

```
/com, Load
```

```
F,6,fy,-6000
```

```
F,6,fx,-Pi
```

```
/com, Considering nonlinear
```

```
/com, effects
```

```
NLGEOM,on
```

```
/com, Load step identification
```

```
/com, time
```

```
TIME,1
```

```
/com, Automatic substep control
```

```
AUTOTS,on
```

```
/com, Limits of substep control
```

```
NSUBST,10,100,5
```

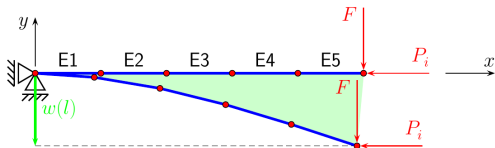
```
/com, Solving the load step
```

```
SOLVE
```

#### Large deformation

Bending moment:

$$M(x, w) = -F(l - x) - P_i[w(l) - w(x)]$$



# Additional ANSYS APDL Commands

## Cantilever with End Load

### Pre Processing

#### Do-Loop

**/PREP7**

**\*DO**, Par, IVAL, FVAL, INC

...any APDL code ...

**\*ENDDO**

execute directly preceding command

**\*REPEAT**, NTOT, VINC1, VINC2, VINC3, VINC4, VINC5, VINC6, ... VINC11  
(specified number of times, incrementing any field)

### Post Processing

table of element values

e.g. axial forces, bending moment

**/POST1**

**ETABLE**, Lab, Item, Comp

display element table

as contoured areas along elements

**PLLS**, LabI, LabJ, Fact, KUND

Prints the element table

**PRETAB**, Lab1, Lab2, Lab3, Lab4, Lab5, Lab6, Lab7, Lab8, Lab9

set display parameter to plot elements

with shapes from their section definition

**/ESHAPE**, SCALE, KEY

(number of circumferential sections are controlled by **SECDATA** parameter)

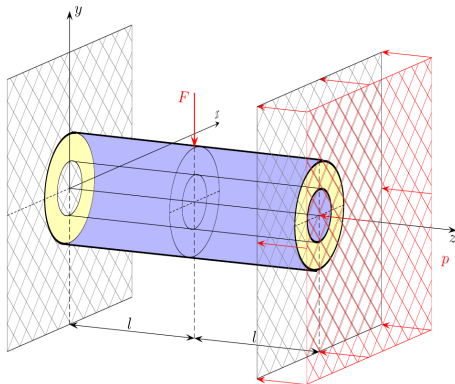
# Results

## Cantilever with End Load

| NODE | $x$  | $u_y$       | $u_y$                    | $u_y$                    | $u_y$                    |
|------|------|-------------|--------------------------|--------------------------|--------------------------|
|      |      | beam theory | $p_0 = 0 \text{ N/mm}^2$ | $p_1 = 2 \text{ N/mm}^2$ | $p_2 = 4 \text{ N/mm}^2$ |
| 1    | 0    | 0.0000      | 0.0000                   | 0.0000                   | 0.0000                   |
| 2    | 200  | -1.2378     | -1.3151                  | -1.4850                  | -1.7167                  |
| 3    | 400  | -4.5978     | -4.7558                  | -5.4165                  | -6.3184                  |
| 4    | 600  | -9.5493     | -9.7902                  | -11.212                  | -13.153                  |
| 5    | 800  | -15.5618    | -15.887                  | -18.262                  | -21.509                  |
| 6    | 1000 | -22.1049    | -22.515                  | -25.945                  | -30.636                  |

# Cantilever with End Load and End Moment

## Problem 2



- Geometric data

$$l = 1000 \text{ mm}$$

$$r_a = 80 \text{ mm}$$

$$r_i = 40 \text{ mm}$$

- Material data (plastics)

$$E = 3000 \text{ N/mm}^2$$

$$\nu = 0.2$$

- Clamping

- Fixed support at both end of the tube
- Pressure applied to wall at one end
- Deformation of cross section possible

- Load (cases 1,2,3)

$$F = 40,000 \text{ N}$$

$$p_0 = 0 \text{ N/mm}^2, p_1 = 4 \text{ N/mm}^2,$$

$$p_2 = 8 \text{ N/mm}^2$$

- Objective target

Bending diagram for  $p_0, p_1, p_2$

Deflection of the tube center

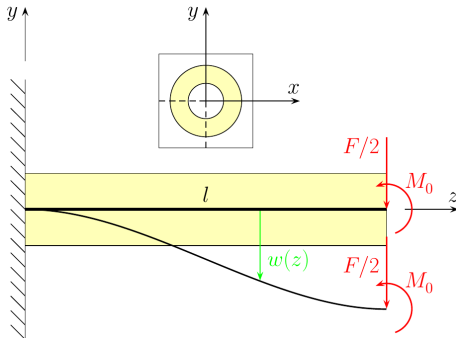
(approximated by displacement of line  
at  $x = r_i, y = 0$ )



# Cantilever with End Load and End Moment

Beam Theory for  $p_0 = 0 \text{ N/mm}^2$

## Reduction to half model



- Moment of inertia

$$I = \frac{\pi}{4}(r_a^4 - r_i^4)$$

- Bending moment

$$M(z) = M_0 - \frac{F}{2} z$$

- Load

$$F = 40,000 \text{ N}$$

## Bending line

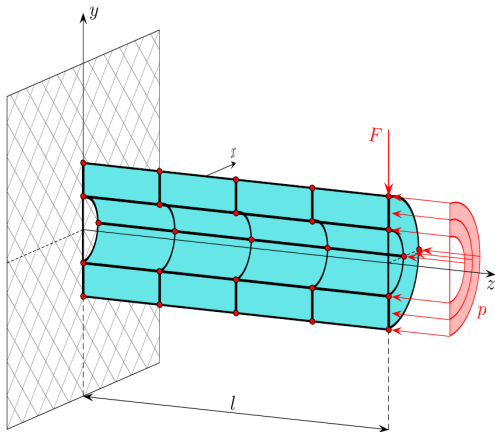
$$w(z) = \frac{F}{6\pi E(r_a^4 - r_i^4)} (3lz^2 - 2z^3)$$

## Deflection of the tube end

$$w(l = 1000 \text{ mm}) = 18.42 \text{ mm}$$

# 3D-Model, Boundary Conditions, Load Situation

Cantilever with End Load and End Moment



- Element types

2-D PLANE182 & 3-D SOLID185

(Simplified enhanced strain formulation)

- Boundary conditions

| Nodes          | Condition     |
|----------------|---------------|
| $x = 0$        | $u_x = 0$     |
| $y = 0, z = 0$ | $u_y = 0$     |
| $z = 0$        | $u_z = 0$     |
| $z = l$        | $u_z$ coupled |

- Modelling and meshing

- Modelling a quarter cross section
- Meshing the quarter area  
(4 elements along arc length,  
2 elements along tube wall thickness)
- Mirroring to a half tube cross section
- Connecting double nodes
- Extrusion to a 3D half model with  
20 elements along the tube length
- Removing the elements of the base  
area ( $x = 0$ )

# Additional ANSYS APDL Commands

## Cantilever with End Load

### Pre Processing

**/PREP7**

Generation circular ring area **CYL4**,XCENTER,YCENTER,RAD1,THETA1,RAD2,THETA2,DEPTH

Mirroring area **ARSYM**,Ncomp,NA1,NA2,NINC,KINC,NOELEM,IMOVE

Compressing objects **NUMMRG**,Label,TOLER,GTOLER>Action,Switch

Adjust numbering **NUMCMP**,label

### Solution

**/SOLU**

coupling DOFs **CP**,NSET,Lab,NODE1,NODE2,NODE3,NODE4,NODE5,NODE6,NODE7,...NODE17

surface load **SFA**,AREA,LKEY,Lab,VALUE,VALUE2

transferring surface load to FE **SFTRAN**

# Results

| NODE | $z$  | Cantilever with End Load |                                   |                                   |                                   |
|------|------|--------------------------|-----------------------------------|-----------------------------------|-----------------------------------|
|      |      | $u_y$<br>beam theory     | $u_y$<br>$p_0 = 0 \text{ N/mm}^2$ | $u_y$<br>$p_1 = 4 \text{ N/mm}^2$ | $u_y$<br>$p_2 = 8 \text{ N/mm}^2$ |
| 8    | 0    | 0.0000                   | 0.0000                            | 0.0000                            | 0.0000                            |
| 100  |      |                          | -0.3061                           | -0.3170                           | -0.3299                           |
| 101  |      |                          | -0.7809                           | -0.8199                           | -0.8652                           |
| 102  |      |                          | -1.5059                           | -1.5909                           | -1.6897                           |
| 103  | 200  | -1.915                   | -2.4292                           | -2.5766                           | -2.7480                           |
| 104  |      |                          | -3.5273                           | -3.7519                           | -4.0131                           |
| 105  |      |                          | -4.7701                           | -5.0849                           | -5.4508                           |
| 106  |      |                          | -6.1287                           | -6.5442                           | -7.0273                           |
| 107  | 400  | -6.484                   | -7.5741                           | -8.0983                           | -8.7081                           |
| 108  |      |                          | -9.0770                           | -9.7155                           | -10.458                           |
| 109  |      |                          | -10.609                           | -11.364                           | -12.243                           |
| 110  |      |                          | -12.140                           | -13.012                           | -14.028                           |
| 111  | 600  | -11.9366                 | -13.641                           | -14.628                           | -15.777                           |
| 112  |      |                          | -15.084                           | -16.180                           | -17.455                           |
| 113  |      |                          | -16.440                           | -17.636                           | -19.029                           |
| 114  |      |                          | -17.678                           | -18.965                           | -20.462                           |
| 115  | 800  | -16.5050                 | -18.771                           | -20.135                           | -21.722                           |
| 116  |      |                          | -19.690                           | -21.116                           | -22.776                           |
| 117  |      |                          | -20.407                           | -21.879                           | -23.592                           |
| 118  |      |                          | -20.884                           | -22.385                           | -24.131                           |
| 35   | 1000 | -18.4207                 | -21.057                           | -22.567                           | -24.324                           |

